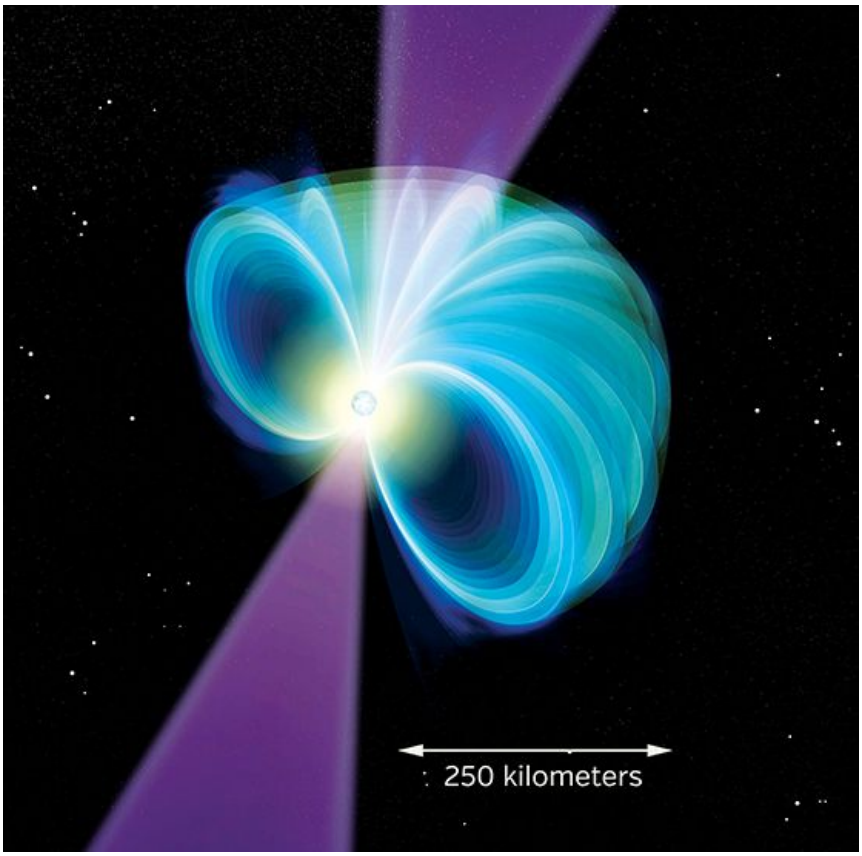


Fig. 2.10. Artist's conception of a neutron star with its donut-shaped magnetic field and its jets.

Neutron stars have very strong magnetic fields, whose force lines are donut-shaped, like the Earth's. Fast-moving particles trapped in a neutron star's magnetic field light up the force lines, producing the blue rings in Figure 2.10. Some of the particles are liberated and stream out the field's poles, producing the two violet jets in the figure. These jets consist of all types of radiation: gamma rays, X-rays; ultraviolet, visual, infrared, and radio waves. As the star spins, its luminous jets sweep around the sky above the neutron star, like a searchlight. Every time a jet sweeps over the Earth, astronomers see a pulse of radiation, so astronomers have named these objects "pulsars."

The universe contains other kinds of fields (collections of force lines) in addition to magnetic fields. One example is electric fields (collections of electric force lines that, for example, drive electric